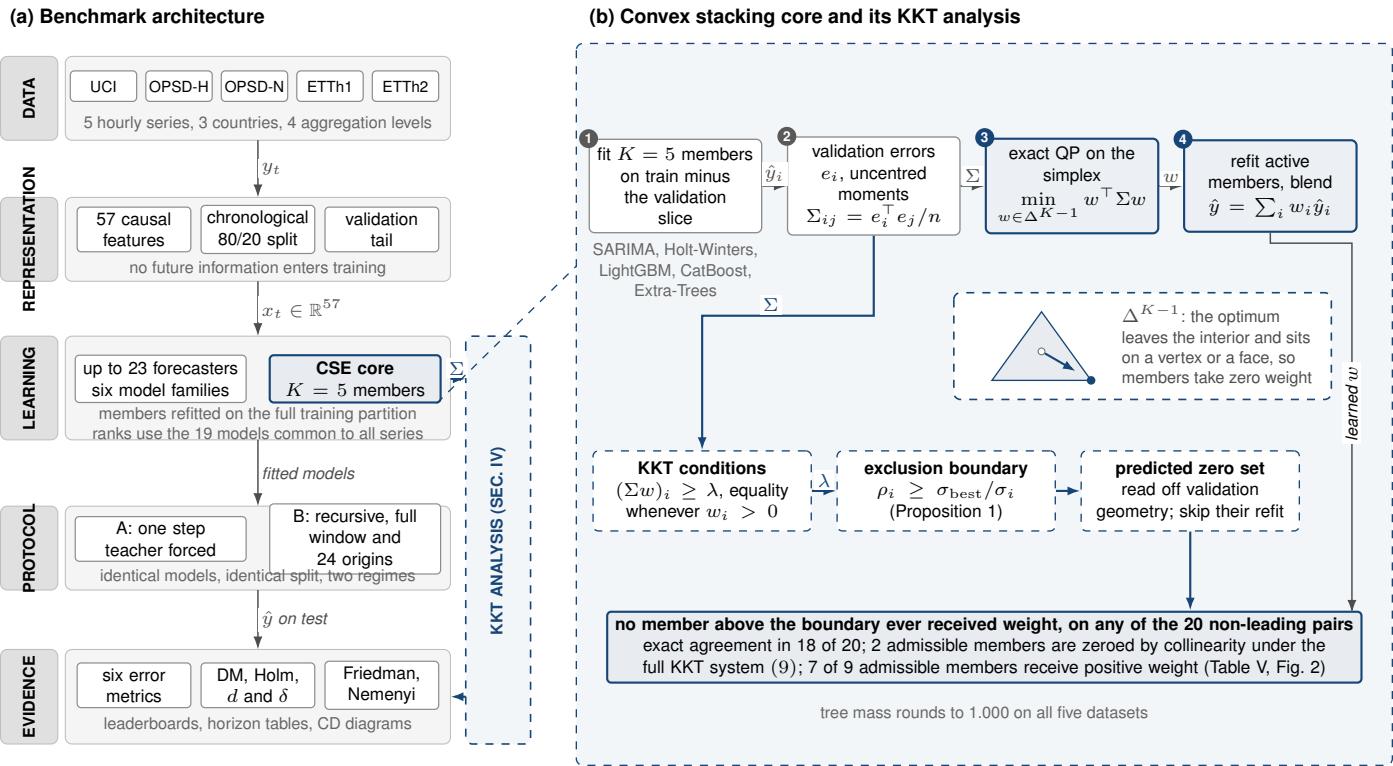

(a) Benchmark architecture
DATA
UCI
OPSD-H
OPSD-N
ETTh1
ETTh2
5 hourly series, 3 countries, 4 aggregation levels
y_t
REPRESENTATION
57 causal features
chronological 80/20 split
validation tail
no future information enters training
$x_t \in \mathbb{R}^{57}$
LEARNING
up to 23 forecasters six model families
CSE core $K = 5$ members
Σ
members refitted on the full training partition
ranks use the 19 models common to all series
fitted models
PROTOCOL
A: one step teacher forced
B: recursive, full window and 24 origins
identical models, identical split, two regimes
$\hat{y}$ on test
EVIDENCE
six error metrics
DM, Holm, d and δ
Friedman, Nemenyi
leaderboards, horizon tables, CD diagrams
KKT ANALYSIS (SEC. IV)
(b) Convex stacking core and its KKT analysis
1 fit $K = 5$ members on train minus the validation slice
SARIMA, Holt-Winters, LightGBM, CatBoost, Extra-Trees
$\hat{y}_i$
2 validation errors e_i, uncentred moments $\Sigma_{ij} = e_i^\top e_j / n$
Σ
3 exact QP on the simplex $\min_{w \in \Delta^{K-1}} w^\top \Sigma w$
w
4 refit active members, blend $\hat{y} = \sum_i w_i \hat{y}_i$
Σ
Δ^{K-1}: the optimum leaves the interior and sits on a vertex or a face, so members take zero weight
learned w
KKT conditions $(\Sigma w)_i \geq \lambda$, equality whenever $w_i > 0$
λ
exclusion boundary $\rho_i \geq \sigma_{\text{best}} / \sigma_i$ (Proposition 1)
predicted zero set read off validation geometry; skip their refit
no member above the boundary ever received weight, on any of the 20 non-leading pairs
exact agreement in 18 of 20; 2 admissible members are zeroed by collinearity under the full KKT system (9); 7 of 9 admissible members receive positive weight (Table V, Fig. 2)
tree mass rounds to 1.000 on all five datasets